# Introduction to Deluge Scripting: Writing Custom Functions and Basic Programs

Case Study 14

## Aim

To understand the basics of Deluge, the scripting language of Zoho Creator, and to write custom functions and simple programs.

## Objective

- Learn Deluge syntax, data types and operators.
- Use conditions, loops and built-in functions.
- Write and call custom functions.
- Work with form data: fetch, insert and update records, and send e-mails.

## About Deluge

**Deluge** (Data Enriched Language for the Universal Grid Environment) is Zoho's online scripting language, used in Zoho Creator, CRM and other Zoho products. It is simple, English-like and designed for automating workflows, validating data and integrating applications. Scripts are written in the Deluge editor (script builder) in workflows, form events, buttons and custom functions.

## Basics of Deluge

| Concept | Details | Example |
|---|---|---|
| Comment | Single-line comment | // this is a comment |
| Variables | Declared by assignment; no type keyword | name = "Ravi"; |
| Data types | Text, Number, Decimal, Boolean, Date, List, Map | marks = 85; passed = true; |
| Operators | + - * / %, == != > <, && \|\| | total = a + b; |
| Output | `info` displays a value (debugging) | info total; |
| List / Map | Collections | {1,2,3} and {"a":1} |
| Built-in system variables | Current date, logged-in user | zoho.currentdate, zoho.loginuser |

## Program 1: Hello and Basic Arithmetic

```
name = "Zoho";
info "Hello " + name;

a = 10;
b = 5;
info "Sum = " + (a + b);
info "Product = " + (a * b);
```

**Output:** Hello Zoho, Sum = 15, Product = 50.

## Program 2: Conditional Statement (Grade Finder)

```
marks = 82;
if(marks >= 90)
{
    grade = "A";
}
else if(marks >= 75)
{
    grade = "B";
}
else if(marks >= 50)
{
    grade = "C";
}
else
{
    grade = "Fail";
}
info "Grade: " + grade;
```

**Output:** Grade: B.

## Program 3: Loop Over a List

```
numbers = {10, 20, 30, 40};
total = 0;
for each n in numbers
{
    total = total + n;
}
info "Total = " + total;
```

**Output:** Total = 100.

## Custom Functions

A custom function is a reusable block of Deluge code with a name, parameters and a return type. In Zoho Creator, functions are created under **Workflow** → **Functions** → **Create New Function**, and can be called from workflows, buttons and other scripts.

```
// Function 1: returns the grade for given marks
string getGrade(int marks)
{
    grade = "Fail";
    if(marks >= 90)
    {
        grade = "A";
    }
    else if(marks >= 75)
    {
        grade = "B";
    }
    else if(marks >= 50)
    {
        grade = "C";
    }
    return grade;
}

// Function 2: calculates total price with 18% tax
decimal calculateTotal(decimal price, int qty)
{
    subtotal = price * qty;
    tax = subtotal * 0.18;
    return subtotal + tax;
}
```

**Calling the functions:**

```
result = thisapp.getGrade(82);
info result;                          // B
total = thisapp.calculateTotal(500, 3);
info total;                           // 1770.0
```

## Working with Form Data

| Task | Deluge Code |
|---|---|
| Fetch records | `students = Student_Registration[Course == "Python"];` |
| Loop through records | `for each s in students { info s.Student_Name; }` |
| Count records | `count = Student_Registration[Course == "Python"].count();` |
| Insert a record | `insert into Student_Registration [ Student_Name = "Ravi" Course = "Python" Added_User = zoho.loginuser ];` |
| Update a record | `rec = Student_Registration[ID == input.ID]; rec.Course = "Data Analytics";` |
| Send e-mail | `sendmail [ from: zoho.adminuserid to: input.Email subject: "Welcome" message: "Thank you for registering" ]` |

## Program 4: Validation on Form Submit

```
// Form workflow: On validate
if(input.Phone_Number.length() != 10)
{
    alert "Phone number must be 10 digits";
    cancel submit;
}
```

## Common Deluge Built-in Functions

| Category | Examples |
|---|---|
| Text | length(), toUpperCase(), toLowerCase(), contains(), replaceAll(), subString() |
| Number | round(), abs(), ceil(), floor() |
| Date | addDay(), addMonth(), daysBetween(), toString() |
| Collection | add(), size(), get(), sum(), keys() |

Syntax shown follows common Deluge usage; verify function names in the Zoho Deluge documentation for the version being used. Run each program in the Zoho Creator script editor and record the actual output.

## Observations

- Deluge needs no type declaration for local variables, but functions require typed parameters and return type.
- Fetching records from a form uses a simple criteria-based syntax.
- Custom functions promote reuse and keep workflow scripts short.
- Validation scripts improve data quality before records are saved.

## Results

- Basic programs using variables, conditions and loops were written.
- Reusable custom functions for grade calculation and price with tax were created.
- Deluge was used for data validation, record handling and e-mail notification.

## Conclusion

Deluge extends no-code Zoho Creator applications with custom logic. Understanding its basic syntax, functions and record-handling statements allows developers to automate complex business rules and integrate applications efficiently.